

Retail Sales Analytics Dashboard Business Intelligence Case Study

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Role: Power BI Developer | Aspiring Data Analyst

This case study documents the design, implementation, and business value of a four-page interactive Power BI dashboard built using the Sample Superstore retail dataset.

1. Executive Summary

Project Background

Retail organizations generate large volumes of transactional data. This project demonstrates how Power BI converts raw data into actionable insights for executives through KPI-driven dashboards.

Business Problem

- Limited visibility into sales and profitability.
- Difficulty identifying top products and customers.
- No centralized regional performance reporting.
- Operational metrics were not easily monitored.

Business Objectives

- Monitor business performance.
- Analyze products, customers and regions.
- Support data-driven decisions.
- Provide an executive reporting solution.

Technology Stack

- Power BI Desktop
- Power Query
- DAX
- Excel
- GitHub

2. Dataset & Data Preparation

Dataset

Attribute	Value
Dataset	Sample Superstore
Domain	Retail
Records	~10,000
Format	Excel
Fact Table	Orders

Data Preparation

- Validated data types
- Removed duplicates
- Created Calendar fields
- Created MonthYear and QuarterYear
- Built relationships
- Created DAX measures
- Optimized report layout

Development Workflow

Raw Data → Power Query → Data Cleaning → Data Model → DAX Measures → Dashboard

Insert Power BI Data Model screenshot here.

3. Dashboard Solution

Executive Business Overview

Provides an executive summary using KPIs, trends, category and customer analysis.

Insert dashboard screenshot here.

Product Intelligence

Analyzes product profitability, category and sub-category performance.

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Customer Intelligence

Evaluates customer value, repeat purchases and profitability.

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Regional Performance

Measures regional sales, shipping efficiency and geographic profitability.

Insert dashboard screenshot here.

4. Dashboard Deep Dive

Executive Overview

Business Question: How is the business performing overall?

- KPIs selected to support decision making.
- Interactive slicers enable self-service analysis.
- Visuals chosen to maximize executive readability.

Product Intelligence

Business Question: Which products drive revenue and profit?

- KPIs selected to support decision making.
- Interactive slicers enable self-service analysis.
- Visuals chosen to maximize executive readability.

Customer Intelligence

Business Question: Who are the most valuable customers?

- KPIs selected to support decision making.
- Interactive slicers enable self-service analysis.
- Visuals chosen to maximize executive readability.

Regional Performance

Business Question: Which regions perform best and where can operations improve?

- KPIs selected to support decision making.
- Interactive slicers enable self-service analysis.
- Visuals chosen to maximize executive readability.

5. Key Business Insights

Sales

- West region generated the highest sales.
- Sales trends remained stable across reporting periods.

Products

- Technology delivered strong profitability.
- Discounted products require pricing review.

Customers

- Top customers contributed a significant share of revenue.
- Segment analysis highlighted profitable customer groups.

Regions

- Regional profitability varied by geography.
- Shipping performance remained relatively stable.

6. Business Recommendations

- Increase investment in high-performing products and regions.
- Review discount policies for low-margin products.
- Develop loyalty programs for high-value customers.
- Optimize logistics in regions with longer shipping times.
- Monitor KPIs through scheduled executive reviews.

Future Enhancements

- Drill-through pages
- Report tooltips
- Row-Level Security (RLS)
- Incremental Refresh
- Power BI Service deployment
- Mobile layout

7. Technical Appendix

Skills Demonstrated

- Power BI Dashboard Development
- Power Query
- DAX
- Data Modeling
- KPI Design
- Interactive Navigation
- Business Storytelling
- Data Visualization

Key Measures

- Total Sales
- Total Profit
- Profit Margin %
- Total Orders
- Total Customers
- Average Order Value
- Repeat Purchase Rate
- Average Shipping Days

Project Outcome

Developed a four-page interactive Power BI solution that transforms retail sales data into actionable business intelligence. The dashboard supports executive reporting through intuitive visualizations, consistent design, KPI tracking, and analytical storytelling.

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